

TRAINING INFORMATION BULLETIN

DATE: April 2020 (Rev. A)

SUBJECT: 4.5L Gaseous Engine Diagnostics

SCOPE: All 4.5L Gaseous Protector Product

PURPOSE: To provide information on troubleshooting the 4.5L gaseous engine.

WARRANTY: Information Only

4.5L Electronic Diagnostics

The Generac 4.5 Liter engine is a 4 cylinder, internal combustion, spark-ignited engine. The engine is controlled by an engine control module (ECM). The ECM controls fuel flow, air flow and ignition timing to maintain optimal engine performance. It does this by reading values from multiple sensors located on various engine components, interpreting the data and adjusting engine fuel flow, timing and intake air flow to maintain proper speed, performance and emissions characteristics.

The ECM is located on the right side of the engine block when looking at the engine from the rear end. A large single connector connects the ECM to the engine harness and also to the generator main control panel. Communication to the main control panel is achieved via a digital communications link CANBUS 2.0 J1939.

Sensors are used to monitor engine performance:

TMAP	Manifold absolute pressure and temperature
CRKS	Crankshaft position
CAMS	Camshaft position
WTS	Engine coolant temperature
O2	Oxygen sensor
TP	Throttle position (part of the ETC connector)
TPS	Mixer position Sensor
AFPS	Air filter pressure sensor
FPS	Fuel pressure sensor
OTS	Oil temperature sensor
LOS	Low oil pressure switch
WLS	Coolant level sensor
CT	Current transformer

Engine performance and speed are controlled by:

ETC	Electronic throttle control valve
EMC	Electronic mixer control valve
IC	Ignition coil

Other control devices:

IR	Ignition relay
DPR	Driver power relay
CAC	Charge air cooler
FS	Fuel solenoid

Theory of operation of ECM ON/OFF

Anytime the engine is commanded to start or an ECM parameter is to be displayed on the controller, the genset controller will first turn on the ECM. The ECM is switched ON by Wire 172 (greater than 6 VDC = ON, less than 6 VDC = OFF). At the same time a CAN bus signal is sent from the control panel to the ECM to go active.

After the ECM is switched ON, it performs internal checks, and within 0.1–0.5 seconds it turns on both the driver power and ignition power relay. The ECM will not allow the engine to start unless it also sees the genset controller command start from CAN message GC1 (Generator Controller 1) "START ENGINE".

When the ECM senses a problem or a malfunction within the engine, a fault code is triggered. The ECM fault code will be displayed on the genset controller.

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When the genset controller commands the engine to stop, it will first send GC1 "STOP" to the ECM. When the ECM can confirm the engine has stopped, the genset controller will turn off ECM via Wire 172. If Wire 172 turns off at any time while the engine is running, but no GC1 message is received, the ECM will treat this as a request to stop the engine. Both Wire 172 and GC1 are needed to start the engine, but either one can stop the engine.

Troubleshooting Sensors

The 4.5L engine uses two different sensor types:

Two Wire Sensors	Three Wire Sensors
CRKS – Crankshaft position	TMAP – Manifold absolute pressure and temperature
WTS – Engine coolant temperature	CAMS – Cam shaft position
O2 – Oxygen sensor	TP – Throttle position (part of ETC connector)
OTS – Oil temperature sensor	TPS – Mixer position sensor
LOS – Low oil pressure switch	AFPS – Air filter pressure sensor
	FPS – Fuel pressure sensor

Troubleshooting Basics

Prior to troubleshooting steps, perform a thorough wiring harness inspection, looking for chafed or exposed conductors or pinched wires. During troubleshooting it will be necessary to make the ECM go active. To do this go to Programming (Dealer level access, Level 2 access code), go to DISPLAY, then scroll down to Engine Hours or ECM version. This will turn on the ECM for a few seconds to display active readings.

Level II Pass Code

ESC Up Up Down Down ESC ESC

NOTE: The display will only be active momentarily (a few seconds). Be prepared to observe and record the reading while displayed.

The 4.5L engine sensors use a weather tight connector. Use of standard test probes will damage the sensor or connector. Use only needle type test probes that fit between the wire and insulator.



Figure 1. Needle Type Test Probes

NOTE: It is not recommended to use any testing device other than the manufacturer's approved test lead adapters (P/N A0000683522).

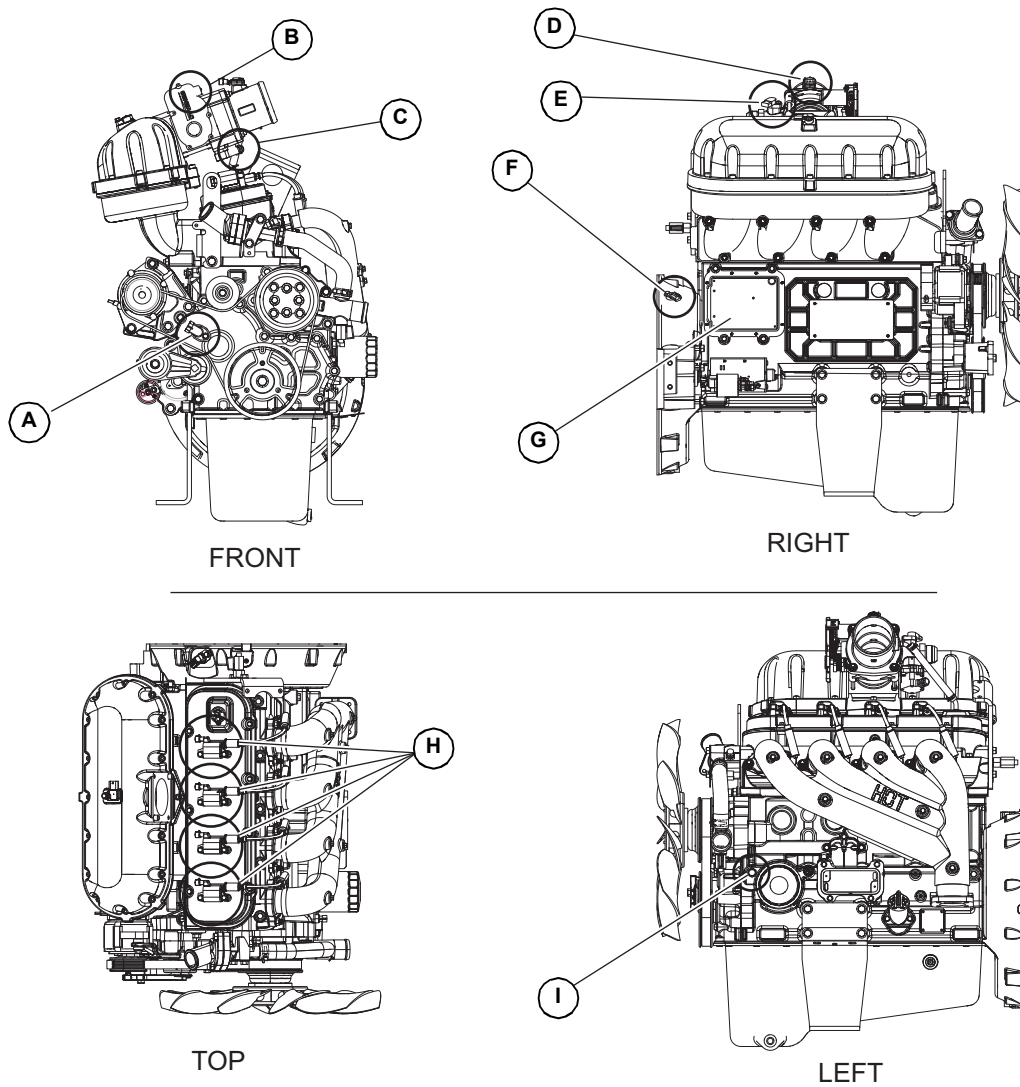


Figure 2. Engine Sensor Locations

A. Camshaft Sensor (CAMS)	F. Crank Sender (CRKS)
B. Electronic Throttle Control (ETC)	G. Engine Control Module (ECM)
C. Electronic Mixer Control (EMC)	H. Ignition Coils (IC1-4)
D. Air Filter Pressure Sensor (AFPS)	I. Oil Temperature Sender (OTS)
E. Manifold Temperature Pressure Sensor (TMAP)	

Troubleshooting Temperature Senders (Fault codes 3700–3717 and 3900–3917)

Sensors such as the OTS and WTS can be unplugged from the harness and tested for resistance.

Ambient Temperature	Resistance
0 °C - 32 °F	9,399 ohms $\pm$ 10%
25 °C - 77 °F	2,795 ohms $\pm$ 10%
50 °C - 122 °F	980 ohms $\pm$ 10%
100 °C - 212 °F	178 ohms $\pm$ 10%

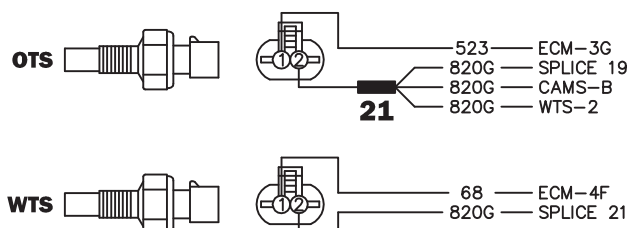


Figure 3. Temperature Senders OTS, WTS

If sensor resistance falls within acceptable parameters disconnect from harness and perform the following tests to verify harness integrity.

1. Use appropriate test probes to test Wire 820G to ground. A reading of 0 ohms or very low ohms should be measured.
2. Use appropriate test probes to test Wire 523 (OTS) or Wire 68 (WTS) for voltage to ground. The ECM will need to be active for this test. Measured voltage should be 4.5–5.0 VDC to ground with an active ECM.

If ground is not measured, or 4.5 VDC is not measured, the harness will need to be tested for continuity of these circuits to the ECM. Special care must be used when working with the ECM connector. Use the wiring diagram for point to point resistance readings of these circuits.

Troubleshooting Crank Sender CRKS (Fault codes 4100–4104)

The crankshaft sensor CRKS can be unplugged from the harness and tested for resistance.

Coil Resistance	330–390 ohms
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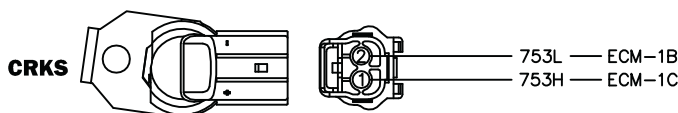


Figure 4. Crank Sender CRKS

Use needle type back probes to test the sensor for cranking voltage and cranking frequency (Hz).

While back probing the sensor wires 753H + Hz signal, 753L - Hz signal

210–270 RPM	270 Hz cranking	1.7 VAC while cranking
1800 RPM	1.7 kHz running	6.5 VAC while running

If the sensor does not read correct output or resistance replace the sensor. If sensor reads the correct output or resistance perform a point to point resistance check on the harness. Verify no continuity to ground on either conductor. Replace as needed.

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Troubleshooting Camshaft Sensor CAMS (Fault codes 4200–4204)

IMPORTANT NOTE: Due to the close proximity of the camshaft sensor to rotating assemblies on the front of the engine, special care must be taken when performing diagnostics.

The camshaft sensor is a three wire sensor. The sensor has a dedicated ground Wire 820G, a dedicated power supply Wire 820B (4.5–5 VDC) and an output Wire 450.

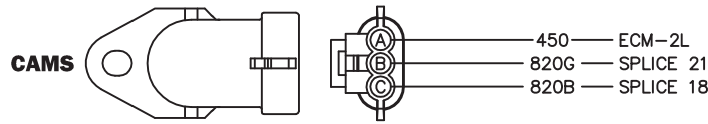


Figure 5. Camshaft Sensor CRKS

In the event of a fault, test the sensor using the following method.

1. Use needle type back probes to verify Wire 820G is grounded to the main ground point on the engine (where the negative battery cable is bolted to the side of the engine block).
 - If Wire 820G is not grounded, test and inspect the harness for breaks and inspect the ECM plug for damaged pins.
2. Test Wire 820B to ground for 4.5–5 VDC when the ECM is active. To make the ECM go active go into programming (Dealer level access, Level 2 access code), go to DISPLAY, then scroll down to Engine Hours or ECM version. This will turn on the ECM for a few seconds to display the readings.
 - If wire 820B does not have voltage present when the ECM is active, test and inspect the harness for breaks and inspect the ECM plug for damaged pins.
3. Test Wire 450 to ground for output voltage when the ECM is active. To make the ECM go active go into programming (Dealer level access, Level 2 access code), go to DISPLAY, then scroll down to engine hours or ECM version. This will turn on the ECM for a few seconds to display the readings. A reading of 4.5–5 VDC should be measured.
 - If no or low voltage is measured remove and inspect the sensor. Replace as necessary.

Troubleshooting Air Filter Pressure Sensor AFPS (Fault codes 3600–3604)

The Air Fuel Pressure Sensor is a 3 wire sensor. The sensor has a dedicated ground Wire 820G, a dedicated power supply Wire 820A (4.5–5 VDC), and an output Wire 604.

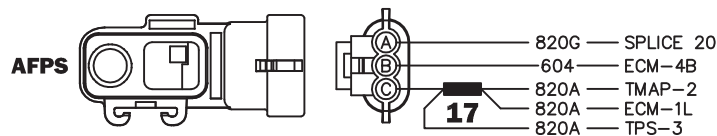


Figure 6. Air Filter Pressure Sensor AFPS

In the event of a fault, test the sensor using the following method:

1. Use needle type back probes to verify Wire 820G is grounded to the main ground point on the engine (where the negative battery cable is bolted to the side of the engine block).
 - If Wire 820G is not grounded, test and inspect the harness for breaks and inspect the ECM plug for damaged pins.
2. Test wire 820A to ground for 4.5–5 VDC when the ECM is active. To make the ECM go active go into programming (Dealer level access, Level 2 access code), go to DISPLAY, then scroll down to engine hours or ECM version. This will turn on the ECM for a few seconds to display the readings.
 - If wire 820A does not have voltage present when the ECM is active, test and inspect the harness for breaks and inspect the ECM plug for damaged pins.
3. Test Wire 604 to ground for output voltage when the ECM is active. To make the ECM go active go into programming (Dealer level access, Level 2 access code), go to DISPLAY, then scroll down to engine hours or ECM version. This will turn on the ECM for a few seconds to display the readings. A reading of 0.5 to 4.5 VDC

should be measured. A clean air filter should produce a very low reading of 0.5 VDC. As the filter begins to clog the reading will increase to 5 VDC.

- If no or low voltage is measured remove and inspect the sensor. AFPS voltage should be 0.5 VDC $\pm$ 10% with sensor removed from system and still connected to harness. Replace as necessary.

Troubleshooting Manifold Temperature Pressure Sensor TMAP (Fault codes 3400–3417 and 3500–3517)

The Manifold Temperature Pressure Sensor is a 4 wire sensor. The sensor has a dedicated ground Wire 820G, a dedicated power supply Wire 820A(4.5–5 VDC), and two output wires—Wire 754 for manifold temperature and Wire 752 for manifold pressure.

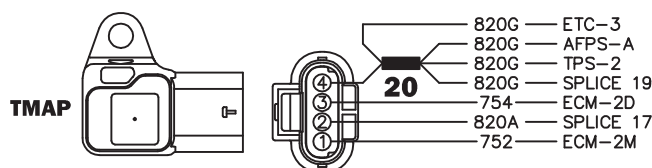


Figure 7. Manifold Temperature Pressure Sensor TMAP

In the event of a fault, test the sensor using the following method:

1. Use needle type back probes to verify Wire 820G is grounded to the main ground point on the engine (where the negative battery cable is bolted to the side of the engine block).
 - If Wire 820G is not grounded, test and inspect the harness for breaks and inspect the ECM plug for damaged pins.
2. Test wire 820A to ground for 4.5–5 VDC when the ECM is active. To make the ECM go active go into programming (Dealer level access, Level 2 access code), go to DISPLAY, then scroll down to engine hours or ECM version. This will turn on the ECM for a few seconds to display the readings.
 - If wire 820A does not have voltage present when the ECM is active, test and inspect the harness for breaks and inspect the ECM plug for damaged pins.
3. Determine which output circuit is faulty, Wire 754 or Wire 752. Test each wire to ground for output voltage when the ECM is active. To make the ECM go active go into programming (Dealer level access, Level 2 access code), go to DISPLAY, then scroll down to engine hours or ECM version. This will turn on the ECM for a few seconds to display the readings. A reading of 0.5–4.5 VDC should be measured.
 - If no or low voltage is measured remove and inspect the sensor. TMAP voltage should be between 2.4–2.6 VDC with sensor removed from system and still connected to harness. Replace as necessary.

Troubleshooting Fuel Pressure sensor FPS (Fault codes 3200–3204)

The Fuel pressure sensor is a 3 wire sensor. The sensor has a dedicated ground Wire 820G, a dedicated power supply Wire (4.5–5 VDC) 820B, and an output Wire 797.

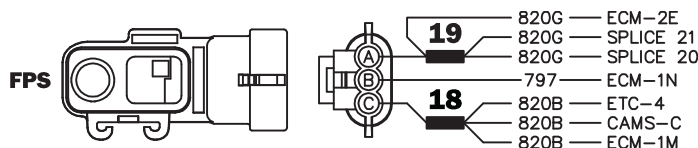


Figure 8. Fuel Pressure Sensor FPS

In the event of a fault, test the sensor using the following method:

1. Use needle type back probes to verify Wire 820G is grounded to the main ground point on the engine (where the negative battery cable is bolted to the side of the engine block).
 - If Wire 820G is not grounded, test and inspect the harness for breaks and inspect the ECM plug for damaged pins.
2. Test wire 820B to ground for 4.5–5 VDC when the ECM is active. To make the ECM go active go into programming (Dealer level access, Level 2 access code), go to DISPLAY, then scroll down to engine hours or ECM version. This will turn on the ECM for a few seconds to display the readings.

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- If wire 820B does not have voltage present when the ECM is active, test and inspect the harness for breaks and inspect the ECM plug for damaged pins.
- 3. Test wire 797 to ground for output voltage when the ECM is active. To make the ECM go active go into programming (Dealer level access, Level 2 access code), go to DISPLAY, then scroll down to engine hours or ECM version. This will turn on the ECM for a few seconds to display the readings. A reading of 0.5–4.5 VDC should be measured.
- If no or low voltage is measured remove and inspect the sensor. FPS voltage should be 1.3 VDC $\pm 10\%$ with sensor removed from system and still connected to harness. Replace as necessary.

Troubleshooting Ignition System Faults (Fault codes 4502–4507 Cylinder 1, 4602–4607 Cylinder 2, 4702–4707 Cylinder 3, 4802–4807 Cylinder 4)

The ignition system on the 4.5L engine is controlled by the ECM. There are four ignition coils (IC 1 through IC 4). The coils are powered by the IR and fuse F4. The IR is controlled by the ECM.

In order to generate a fault the following conditions may apply:

- The ECM will check if ignition coil current is above the threshold, when the coil is firing.
- The ECM will check if ignition coil current is below the threshold, when the coil is firing.
- The ECM will check if ignition coil current is detected when the coils are not firing.

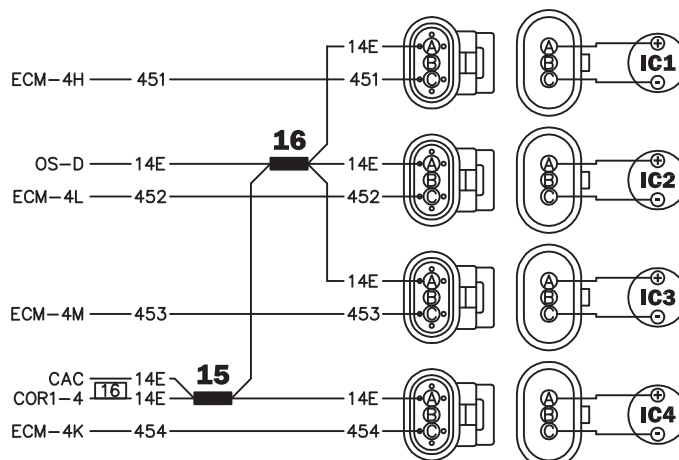


Figure 9. Ignition Coils IC1–IC4

Before troubleshooting is performed check spark plug wires for fit and condition. Check each coil connector for 12 VDC on wire 14E. Check the F4 fuse to see if it is blown or missing or if the relay (IR) is non operational. If the relay is non-functional or the fuse is blown the ignition system will indicate an ignition fault. Check fuse F4 for integrity, and verify relay operation. To verify relay operation you will need to make the ECM go active. To make the ECM go active go into programming (Dealer level access, Level 2 access code), go to DISPLAY, then scroll down to engine hours or ECM version. This will turn on the ECM for a few seconds to display the readings. You can then directly check the output of the relay (IR) either at the relay or anywhere on Wire 14D/14F.

Testing Ignition Coil

To test an ignition coil on the 4.5L engine, disconnect the electrical connection and the primary wire from the coil.

NOTE: It may be necessary to remove the coil from the valve cover.

Primary resistance (measured through coil pins 14E and 454–451)	1.0 Ohm
Secondary resistance (454–451 measured to any primary coil terminal)	9 k Ohm

Troubleshooting Oxygen Sensor OS System Faults (Fault Codes 4300–4304 4315–4317)

The oxygen sensor element, or O₂ sensor, operates by measuring the difference in oxygen between exhaust gas and external air (sample air). A voltage is generated based on the difference between the two. The oxygen sensors output is on Wire 804 and Wire 805. A high voltage (~0.6–0.9 V) indicates a rich condition (low oxygen content) in the exhaust flow, and a low voltage (~0.1–0.4 V) indicates a lean condition (high oxygen content) in the exhaust. Wire 607 and Wire 14E are for the O₂ sensors heater circuit. The heater maintains the O₂ sensor temperature allowing the O₂ sensor to achieve a better resolution.

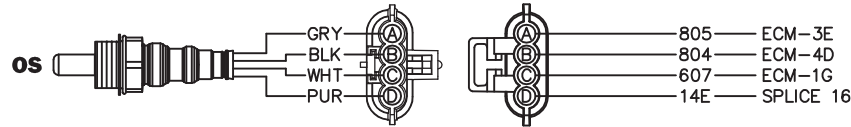


Figure 10. Oxygen Sensor OS

Procedure

1. Locate and disconnect the oxygen sensor. Keep it accessible for the duration of this testing procedure.
2. Set a DMM to measure resistance.
3. Connect one meter test lead to terminal C of the sensor and the other meter test lead to terminal D. Measure and record the resistance.
 - a. If the DMM indicated INFINITY, replace sensor.
 - b. If the DMM indicated approximately 15–21 ohms of resistance, proceed to Step 4.

NOTE: OS resistance at ambient temperature is 15–21 ohms. Resistance may read higher if unit was running prior to troubleshooting.

4. Connect a resistive load bank to the unit.
5. Set the DMM to measure DC voltage.
6. Connect one meter test lead to terminal A of the sensor and connect the other meter test lead to terminal B.
7. Set the AUTO-OFF-MANUAL switch to MANUAL.
8. Once engine is running and stabilized, the generator to greater than 25% of the generators load.
9. With the generator running under load, measure and record the DC voltage indicated by the DMM.
 - a. If the DMM indicated a varying DC voltage between 0.3–0.9 VDC, proceed to Step 11.
 - b. If the DMM did not indicate a varying DC voltage, replace oxygen sensor.

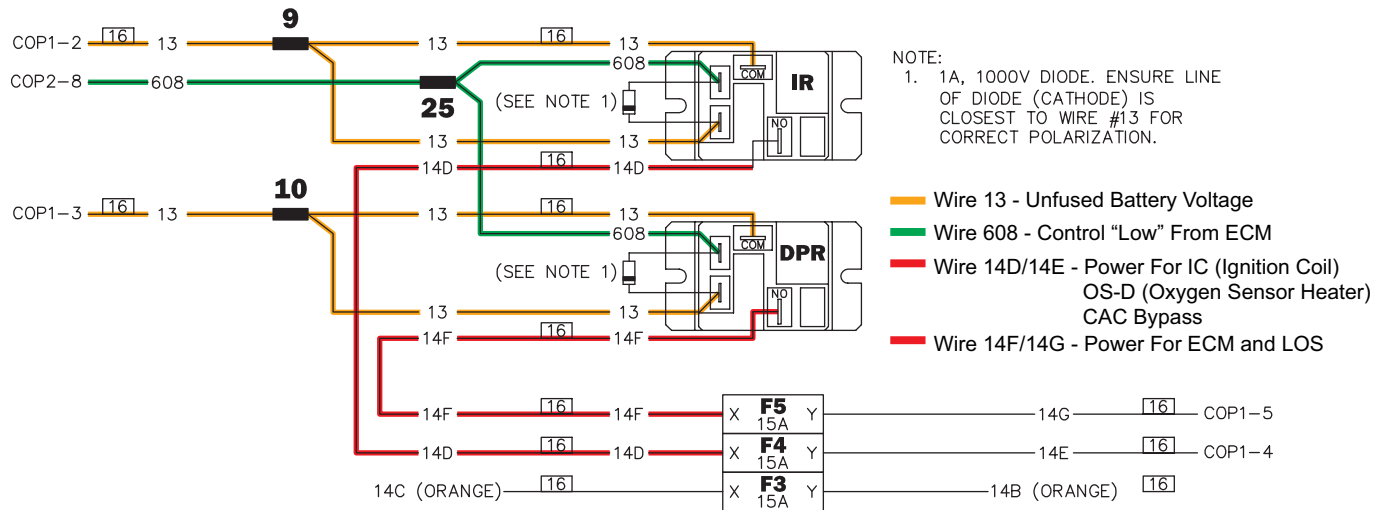
NOTE: Only a short time window exists to test the wiring harness (approximately two minutes after the engine shuts off).

10. Set the AUTO-OFF-MANUAL switch to OFF.
11. Locate and disconnect the oxygen sensor connector from the harness.
12. Set a DMM to measure resistance.
13. Connect one meter test lead to Wire 804 and the other meter test lead to Wire 805. Measure and record the resistance.

NOTE: This step is measuring the resistance of the sensor itself. With the engine shut down resistance will start low (approximately 20 Ohms) and rise to an INFINITE value. A rising resistance indicates the wire integrity is good.

- a. If the DMM did not indicate the proper resistance, repair or replace wiring between the controller.

The ignition relay (IR) and the driver power relay (DPR) are both installed behind the main control panel, on the side bulkhead. They get power (for the coil and for the output) from Wire 13. Both relays are controlled by Wire 608 which is connected to the ECM at terminal 1H. This is a control ground so that when the ECM needs these relays active, the ECM will pull Wire 608 low, energizing the relays. The IR relay provides power for the ignition coils, the oxygen sensor heater, and the CAC bypass. The DPR relay provides power for the controller when ECM is active or running.



Troubleshooting The Mixer Position Sensor (TPS) and Electronic Mixer Control Valve (EMC) (Fault codes 4900–4904, 4907, 4915–4917)

The electronic mixer control (EMC) is positioned on the inlet of the fuel mixer. The EMC receives fuel from two fuel valves located on the frame rail. Fuel inlet pressure to the EMC is 3.5–14 in H₂O. The EMC controls fuel flow to the engine to maintain performance and emissions characteristics. The valve is controlled by a two coil stepper motor. Stepper motor coil resistance is 3.6 ohms.

The mixer position sensor (TPS) is a feedback sensor located on the side of the EMC. The TPS is a 3 wire sensor. The sensor has a dedicated ground Wire 820G, a dedicated power supply Wire (4.5–5 VDC) 820A, and feedback to the ECM Wire 606 (0.5–4.5 VDC)

In the event of a fault code check stepper motor Wire 605A to Wire 605B for resistance (3.6 ohms) then check Wire 605C to Wire 605D for resistance (3.6 ohms).

NOTE: The EMC and the TPS are components of a single assembly. If either component is determined to be faulty the mixer assembly must be replaced.

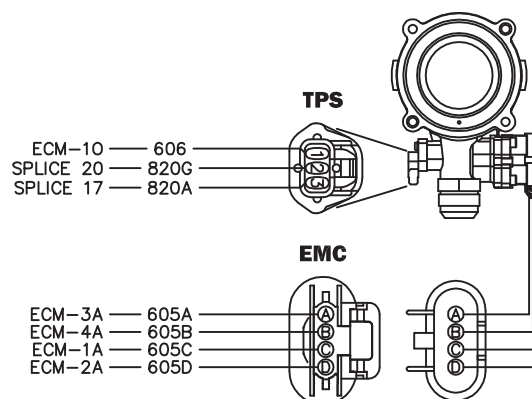


Figure 11. Mixer Position Sensor (TPS) and Electronic Mixer Control Valve (EMC)

Troubleshooting The Electronic Throttle Control Valve (ETC) (Fault codes 3100–3104, 3107, 3115–3117)

The electronic throttle control valve (ETC) is a motor controlled throttle valve with a built in throttle position feedback sensor. A 5 wire harness connects the ETC to the ECM for throttle control and throttle position feedback.

The sensor has a dedicated ground Wire 820G and a dedicated power supply Wire 820B (4.5–5 VDC). Wire 766 is the sensor output to the ECM (4.5 VDC wide open throttle, 0.5 VDC throttle fully closed).

Wire 770 is ETC positive (+), and Wire 771 is ETC negative (–), with a motor resistance of 1.5 ohms. This resistance is measured at the motor drive terminals located inside the cover of the ETC valve. Resistance will be higher (15–35 ohms) if measured on the external cover.

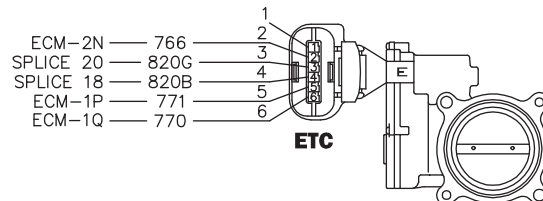


Figure 12. Electronic Throttle Control Valve (ETC)

NOTE: Needle probes are not recommended for the type of connector used on the ETC valve harness. If the control panel generates any ETC codes, Operate the throttle plate manually with a non-metallic probe to check for smooth movement. Replace the ETC valve if necessary.

Problems With Loading and Overspeed Shutdown

If a unit is having a hard time with loading (frequency dips excessive or response time is poor) or when going from a heavy load to a low or no load and the unit overspeeds and shuts down, there may be a problem with the ECM current transformer (CT).

The CT is installed on one of the power leads of alternators output. The CT detects large load swings that are being applied to the unit. As these load swings hit the unit, the ECM will temporarily modify the emission characteristics for better load response.

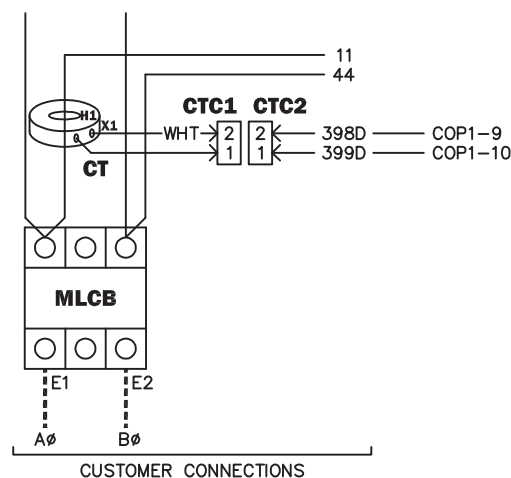


Figure 13. Current Transformer (CT)

Type	Code	Description	Fault
ALARM	3000	Alarm SPN/FMI not supported by genset controller	Hardware Match Error
ALARM	3001	Speed Command Mismatch - ECM not running at commanded speed	Speed Match Error
ALARM	3002	Fuel Command Mismatch - ECM not operating on the commanded Fuel Type	Fuel Match Error
ALARM	3003	ECM initiated Emergency Shutdown but no DTC received by controller	
WARNING	3015	Warning SPN/FMI not supported by genset controller	Hardware Match Error
ALARM	3100	Data valid but above normal operating range; high severe level	Throttle Body Air
ALARM	3101	Data valid but below normal operating range; high severe level	Throttle Body Air
ALARM	3102	Data erratic, intermittent, or incorrect	Throttle Body Air
ALARM	3103	Voltage above normal range; shorted to high source	Throttle Body Air
ALARM	3104	Voltage below normal range; shorted to low source; open circuit	Throttle Body Air
ALARM	3107	Mechanical system not responding correctly	Throttle Body Air
WARNING	3115	Data valid but above normal operating range; least severe level	Throttle Body Air
WARNING	3117	Data valid but below normal operating range; least severe level	Throttle Body Air
ALARM	3200	Data valid but above normal operating range; high severe level	High Fuel Pressure
ALARM	3201	Data valid but below normal operating range; high severe level	Low Fuel Pressure
ALARM	3202	Data erratic, intermittent, or incorrect	Fuel Pressure
ALARM	3203	Voltage above normal range; shorted to high source	Fuel Pressure
ALARM	3204	Voltage below normal range; shorted to low source; open circuit	Fuel Pressure
WARNING	3215	Data valid but above normal operating range; least severe level	High Fuel Pressure
WARNING	3217	Data valid but below normal operating range; least severe level	Low Fuel Pressure
ALARM	3300	Data valid but above normal operating range; high severe level	High Oil Pressure
ALARM	3301	Data valid but below normal operating range; high severe level	Low Oil Pressure
ALARM	3302	Data erratic, intermittent, or incorrect	Oil Pressure
ALARM	3303	Voltage above normal range; shorted to high source	Oil Pressure
ALARM	3304	Voltage below normal range; shorted to low source; open circuit	Oil Pressure
WARNING	3315	Data valid but above normal operating range; least severe level	High Oil Pressure
WARNING	3317	Data valid but below normal operating range; least severe level	High Oil Pressure
ALARM	3400	Data valid but above normal operating range; high severe level	High MAP
ALARM	3401	Data valid but below normal operating range; high severe level	Low MAP
ALARM	3402	Data erratic, intermittent, or incorrect	MAP
ALARM	3403	Voltage above normal range; shorted to high source	MAP
ALARM	3404	Voltage below normal range; shorted to low source; open circuit	MAP
WARNING	3415	Data valid but above normal operating range; least severe level	High MAP

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Type	Code	Description	Fault
WARNING	3417	Data valid but below normal operating range; least severe level	Low MAP
ALARM	3500	Data valid but above normal operating range; high severe level	High MAP Temp
ALARM	3501	Data valid but below normal operating range; high severe level	Low MAP Temp
ALARM	3502	Data erratic, intermittent, or incorrect	MAP Temp
ALARM	3503	Voltage above normal range; shorted to high source	MAP Temp
ALARM	3504	Voltage below normal range; shorted to low source; open circuit	MAP Temp
WARNING	3515	Data valid but above normal operating range; least severe level	High MAP Temp
WARNING	3517	Data valid but below normal operating range; least severe level	Low MAP Temp
ALARM	3600	Data valid but above normal operating range; high severe level	Check Air Filter
ALARM	3601	Data valid but below normal operating range; high severe level	Check Air Filter
ALARM	3602	Data erratic, intermittent, or incorrect	Air Filter Sensor
ALARM	3603	Voltage above normal range; shorted to high source	Air Filter Sensor
ALARM	3604	Voltage below normal range; shorted to low source; open circuit	Air Filter Sensor
WARNING	3615	Data valid but above normal operating range; least severe level	Check Air Filter
WARNING	3617	Data valid but below normal operating range; least severe level	Check Air Filter
ALARM	3700	Data valid but above normal operating range; high severe level	High Coolant Temp
ALARM	3701	Data valid but below normal operating range; high severe level	Low Coolant Temp
ALARM	3702	Data erratic, intermittent, or incorrect	Coolant Temp
ALARM	3703	Voltage above normal range; shorted to high source	Coolant Temp
ALARM	3704	Voltage below normal range; shorted to low source; open circuit	Coolant Temp
WARNING	3715	Data valid but above normal operating range; least severe level	High Coolant Temp
WARNING	3717	Data valid but below normal operating range; least severe level	Low Coolant Temp
ALARM	3803	Voltage above normal range; shorted to high source	High Battery Voltage
ALARM	3804	Voltage below normal range; shorted to low source; open circuit	Low Battery Voltage
ALARM	3900	Data valid but above normal operating range; high severe level	High Oil Temp
ALARM	3901	Data valid but above normal operating range; high severe level	Low Oil Temp
ALARM	3902	Data valid but below normal operating range; high severe level	Oil Temp
ALARM	3903	Data erratic, intermittent, or incorrect	Oil Temp
ALARM	3904	Voltage above normal range; shorted to high source	Oil Temp
WARNING	3915	Voltage below normal range; shorted to low source; open circuit	High Oil Temp
WARNING	3917	Data valid but below normal operating range; least severe level	High Oil Temp
ALARM	4003	Voltage above normal range; shorted to high source	Fuel System

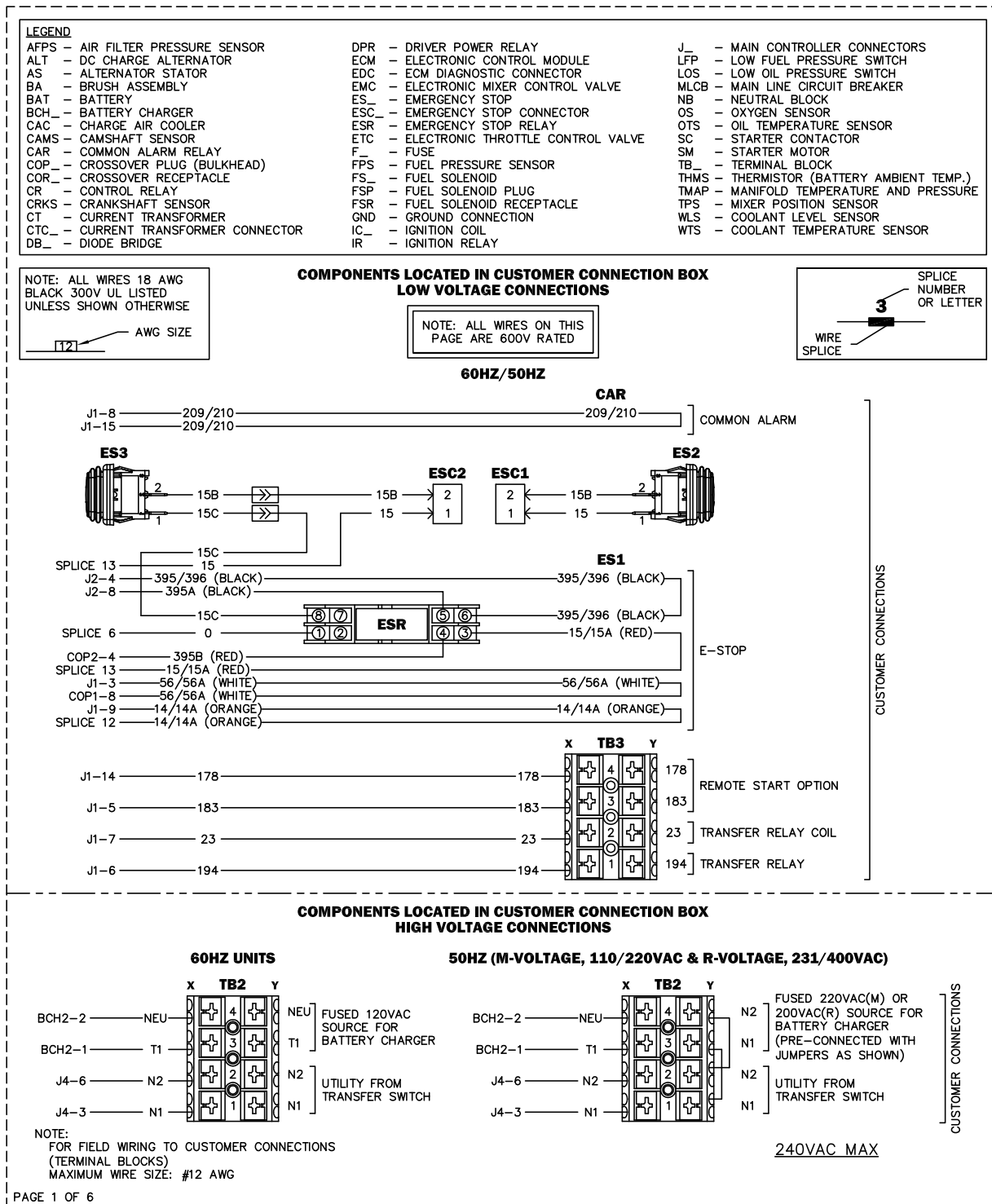
Type	Code	Description	Fault
ALARM	4004	Voltage below normal range; shorted to low source; open circuit	Fuel System
ALARM	4100	Data valid but above normal operating range; high severe level	Crankshaft Sensor
ALARM	4101	Data valid but below normal operating range; high severe level	Crankshaft Sensor
ALARM	4102	Data erratic, intermittent, or incorrect	Crankshaft Sensor
ALARM	4103	Voltage above normal range; shorted to high source	Crankshaft Sensor
ALARM	4104	Voltage below normal range; shorted to low source; open circuit	Crankshaft Sensor
ALARM	4200	Data valid but above normal operating range; high severe level	Camshaft Sensor
ALARM	4201	Data valid but below normal operating range; high severe level	Camshaft Sensor
ALARM	4202	Data erratic, intermittent, or incorrect	Camshaft Sensor
ALARM	4203	Voltage above normal range; shorted to high source	Camshaft Sensor
ALARM	4204	Voltage below normal range; shorted to low source; open circuit	Camshaft Sensor
ALARM	4300	Data valid but above normal operating range; high severe level	Check O2 Sensor
ALARM	4301	Data valid but below normal operating range; high severe level	Check O2 Sensor
ALARM	4302	Data erratic, intermittent, or incorrect	Check O2 Sensor
ALARM	4303	Voltage above normal range; shorted to high source	Check O2 Sensor
ALARM	4304	Voltage below normal range; shorted to low source; open circuit	Check O2 Sensor
WARNING	4315	Data valid but above normal operating range; least severe level	Check O2 Sensor
WARNING	4317	Data valid but below normal operating range; least severe level	Check O2 Sensor
ALARM	4400	Data valid but above normal operating range; high severe level	Engine Knock Sensor
ALARM	4401	Data valid but below normal operating range; high severe level	Engine Knock Sensor
ALARM	4402	Data erratic, intermittent, or incorrect	Engine Knock Sensor
ALARM	4403	Voltage above normal range; shorted to high source	Engine Knock Sensor
ALARM	4404	Voltage below normal range; shorted to low source; open circuit	Engine Knock Sensor
WARNING	4415	Data valid but above normal operating range; least severe level	Engine Knock Sensor
WARNING	4417	Data valid but below normal operating range; least severe level	Engine Knock Sensor
ALARM	4502	Data erratic, intermittent, or incorrect	Ignition Coil 1
ALARM	4503	Voltage above normal range; shorted to high source	Ignition Coil 1
ALARM	4504	Voltage below normal range; shorted to low source; open circuit	Ignition Coil 1
ALARM	4507	Mechanical system not responding correctly	Ignition Coil 1
ALARM	4602	Data erratic, intermittent, or incorrect	Ignition Coil 2
ALARM	4603	Voltage above normal range; shorted to high source	Ignition Coil 2
ALARM	4604	Voltage below normal range; shorted to low source; open circuit	Ignition Coil 2

Confidential information subject to Article IV of the Domestic Non-Exclusive Buy/Sell Agreement. Do not disclose.



Type	Code	Description	Fault
ALARM	4607	Mechanical system not responding correctly	Ignition Coil 2
ALARM	4702	Data erratic, intermittent, or incorrect	Ignition Coil 3
ALARM	4703	Voltage above normal range; shorted to high source	Ignition Coil 3
ALARM	4704	Voltage below normal range; shorted to low source; open circuit	Ignition Coil 3
ALARM	4707	Mechanical system not responding correctly	Ignition Coil 3
ALARM	4802	Data erratic, intermittent, or incorrect	Ignition Coil 4
ALARM	4803	Voltage above normal range; shorted to high source	Ignition Coil 4
ALARM	4804	Voltage below normal range; shorted to low source; open circuit	Ignition Coil 4
ALARM	4807	Mechanical system not responding correctly	Ignition Coil 4
ALARM	4900	Data valid but above normal operating range; high severe level	Fuel Mixer
ALARM	4901	Data valid but below normal operating range; high severe level	Fuel Mixer
ALARM	4902	Data erratic, intermittent, or incorrect	Fuel Mixer
ALARM	4903	Voltage above normal range; shorted to high source	Fuel Mixer
ALARM	4904	Voltage below normal range; shorted to low source; open circuit	Fuel Mixer
ALARM	4907	Mechanical system not responding correctly	Fuel Mixer
WARNING	4915	Data valid but above normal operating range; least severe level	Fuel Mixer
WARNING	4917	Data valid but below normal operating range; least severe level	Fuel Mixer
ALARM	5003	Voltage above normal range; shorted to high source	CAC
ALARM	5004	Voltage below normal range; shorted to low source; open circuit	CAC
ALARM	5100	Data valid but above normal operating range; high severe level	RPM High
ALARM	5101	Data valid but below normal operating range; high severe level	RPM Low
ALARM	5102	Data erratic, intermittent, or incorrect	RPM
ALARM	5107	Mechanical system not responding correctly	RPM
WARNING	5115	Data valid but above normal operating range; least severe level	RPM High
WARNING	5117	Data valid but below normal operating range; least severe level	RPM Low

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COMPONENTS LOCATED IN CUSTOMER CONNECTION BOX HIGH VOLTAGE CONNECTIONS

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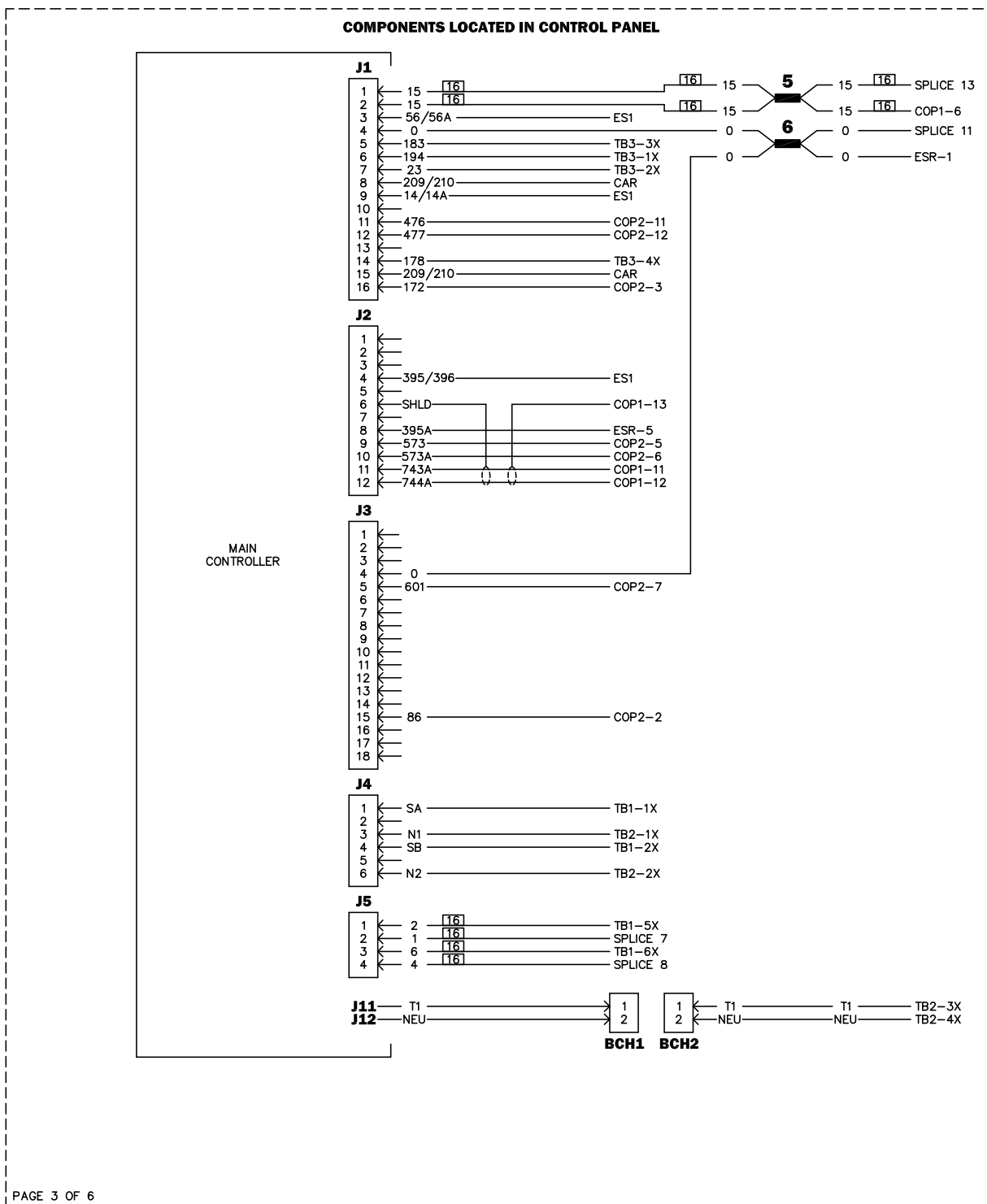
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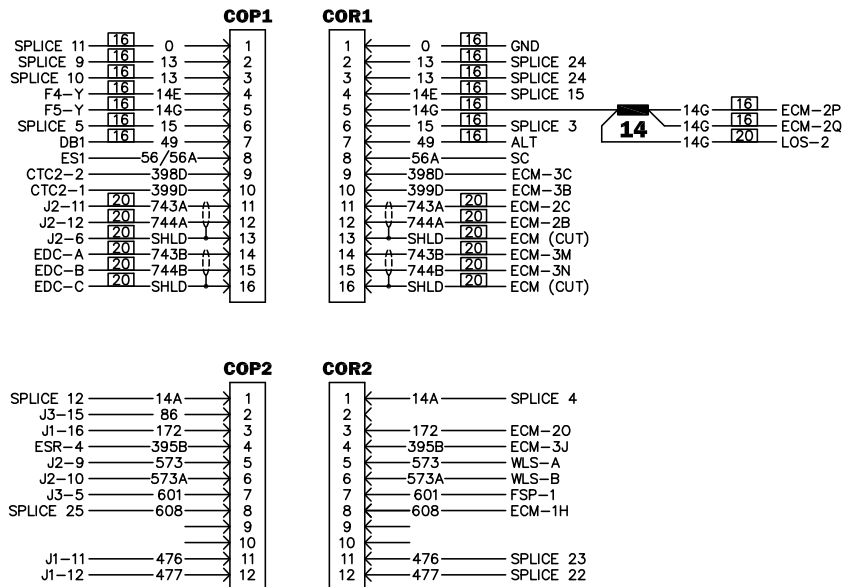
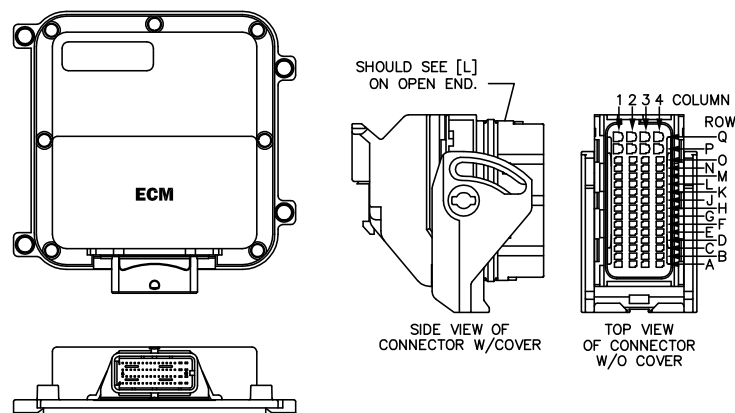
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COMPONENTS LOCATED IN CUSTOMER CONNECTION BOX & ON ENGINE
BULKHEAD CROSSOVER CONNECTIONSCOMPONENTS LOCATED ON ENGINE
ELECTRONIC CONTROL MODULE

PIN	WIRE	TO
1A	605C	EMC-C
1B	753L	CRKS-2
1C	753H	CRKS-1
1D	760	CAC
1E	-	-
1F	-	-
1G	607	OS-C
1H	608	COR2-8
1J	610	FS1-B
1K	-	-
1L	820A	SPLICE 17
1M	820B	SPLICE 18
1N	797	FPS-B
1O	606	TPS-1
1P	771	ETC-5
1Q	770	ETC-6

PIN	WIRE	TO
2A	605D	EMC-D
2B	744A	COR1-12
2C	743A	COR1-11
2D	754	TMAP-3
2E	820G	SPLICE 19
2F	-	-
2G	-	-
2H	-	-
2J	-	-
2K	-	-
2L	450	CAMS-A
2M	752	TMAP-1
2N	766	ETC-2
2O	172	COR2-3
2P	14G	SPLICE 14
2Q	14G	SPLICE 14

PIN	WIRE	TO
3A	605A	EMC-A
3B	399D	COR1-10
3C	398D	COR1-9
3D	-	-
3E	805	OS-A
3F	-	-
3G	523	OTS-1
3H	-	-
3J	395B	COR2-4
3K	-	-
3L	86	LOS-1
3M	743B	COR1-14
3N	744B	COR1-15
3O	15	SPLICE 3
3P	0	GND
3Q	0	GND

PIN	WIRE	TO
4A	605B	EMC-B
4B	604	AFPS-B
4C	-	-
4D	804	OS-B
4E	-	-
4F	68	WTS-1
4G	-	-
4H	451	IC1-C
4J	-	-
4K	454	IC4-C
4L	452	IC2-C
4M	453	IC3-C
4N	-	-
4O	-	-
4P	0	GND
4Q	0	GND

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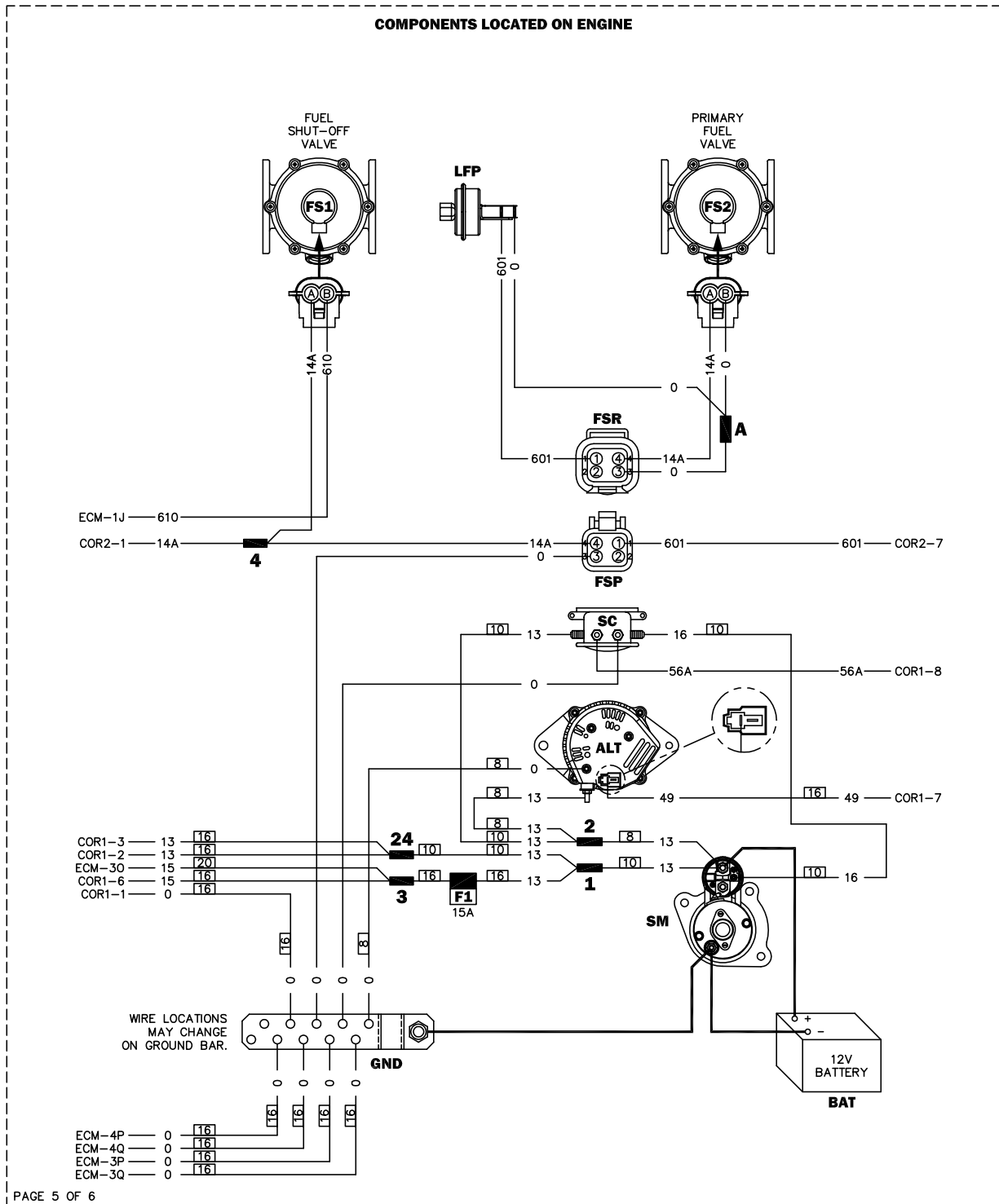
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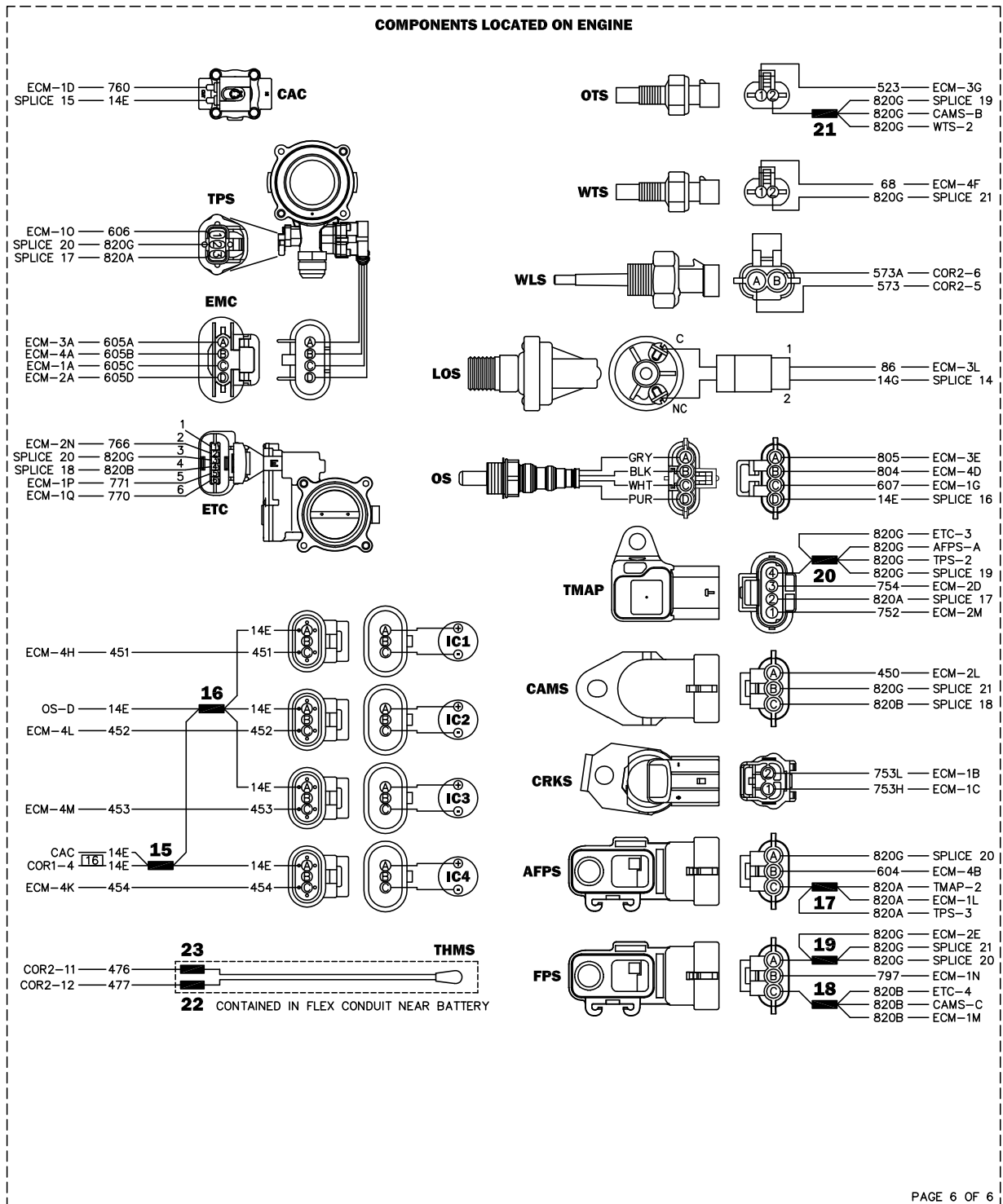
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